The University of the National Football League: How Technology, Injury Surveillance, and Health Care Have Improved the Safety of America's Game

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Abstract

American football has become one of the most popular sports in the United States. Despite the millions of players at all levels of competition who gain the physical, social, and psychological rewards that football provides, many interested stakeholders continue to ask, "Is football safe?" Although there are only approximately 1,700 players on National Football League (NFL) rosters, the injuries they sustain have garnered the most attention—and criticism—from the national media. Increased public awareness of the injury potential football possesses has led to an open debate and a major shift in public sentiment over the past 5 years. Although no sport is perfectly safe, the question is whether it can be made *relatively* safe and if the long-term consequences are worth the risk. This article reviews the methods by which one sports league—the NFL—has used advances in medical technology and injury surveillance to improve the health and safety of its players.

Keywords
- football
- technology
- health care
- injury surveillance
- NFL

Football as "America's Game"

From Pop Warner leagues for children to the explosive action of the National Football League (NFL), football has replaced baseball in the minds of many Americans as the "national pastime." More than 3 million children ages 6 to 14 play organized youth tackle football[1] and another 1.08 million play at the high school level.[2] Despite popular media reports that youth football participation is declining, such as the oft-cited 2013 ESPN report[3] showing a nearly 10% drop in Pop Warner youth football participation over a 2-year period, American tackle football remains the most popular participation and spectator sport in the United States.[4]

With the consolidation of television rights contracts by then commissioner Pete Rozelle, the NFL football surpassed baseball in terms of popularity in the late 1970s.[5] Following America's involvement in the Vietnam War and the "Cold War" with the Soviet Union, the United States was seen as a global superpower as a result of its military. It is no coincidence that, until recently, the game was often depicted with allusions to actual war using such militaristic metaphors as referring to the quarterback as the team's "field general," who threw "bombs" as part of the team's "aerial attack," and proudly stating that linemen are "in the trenches" doing "hand-to-hand combat" to defend against an all-out "blitz" by the defense. What was once a game limited to Sunday afternoons, NFL football is now available up to four nights a week throughout the regular season to capitalize on the public's thirst for this mass media behemoth. The 2015 NFL season opener of "Thursday Night Football" (TNF) shown on 17th September was the most-watched and highest-rated TNF broadcast.[6] That game was seen by more than 21 million viewers and approached viewership of Game 7 of the 2014 World Series, widely considered the most exciting baseball

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finish in recent memory, with 23.5 million viewers.[7] Obviously, those numbers are far shy of the viewership for the 2015 Super Bowl, whose 114.4 million viewers made it the most-watched television show in U.S. history.[8] Although several major sports leagues occupy the top 50 list of social media technology, the NFL remains the brand that is consistently most discussed, currently ranking at number 7 of all brands according to media analyst, Infegy Atlas (Infegy, Kansas City, MO). Accordingly, the NFL remains financially extremely viable, with the total value of all 32 franchises listed at $62.9 billion, an average team value of $1.97 billion and an average operating income of $78.8 million.[9] These values far outpace those of the other three major sports leagues, each of which has 30 teams[10–12] (►**Figs. 1** and **2**). Because of this economic footprint and high degree of visibility, the NFL has been widely scrutinized for its handling and prevention of injuries and, most recently, head trauma. Although injuries are an accepted part of football as a collision sport, several highly publicized cases and an ever increasing popularity of fantasy football and sports betting have led to an intense focus on football injuries and the medical aspects of the game which has intensified a national dialogue on the safety of the sport.

The National Football League Reflects American Society

The NFL can be seen as a microcosm of American society. Political, economic, and social issues affect the league's players just as other American citizens do. There are many positive social benefits experienced by playing professional football such as achieving the American dream of financial security, working toward a unified goal (e.g., the Super Bowl), receiving compensation based on performance, and working in an ethnically diverse workplace. A core principle which has strengthened the league financially in which the wealthier teams profit share with the "poorer" ones can be compared metaphorically with the rich, giving to the poor for the overall greater good. These positives are unfortunately offset by the negative realities of drug and alcohol addiction, domestic violence, gun possession, and cheating that plague not only the NFL but also the rest of modern society. Even controversial issues such as gay rights have touched the NFL with the recent drafting of the first openly gay player—Michael Sam—by the St. Louis Rams. Thus, despite possessing a set of highly unique athletic skills and receiving compensation far above the national average, NFL players are not necessarily shielded from the reality of social and cultural issues that influence all other American adults.

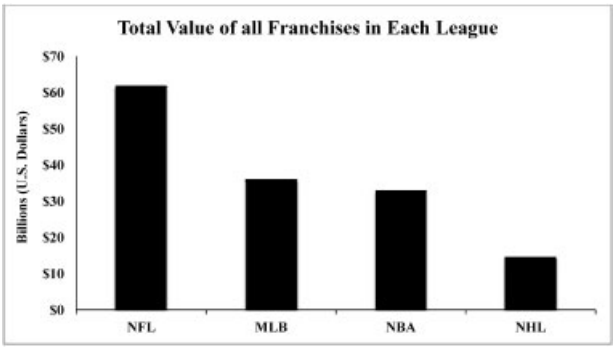

Fig. 1 Value of the four major American sports leagues (in billions of U.S. dollars).[9–12] MLB, Major League Baseball; NBA, National Basketball Association; NFL, National Football League; NHL, National Hockey League.

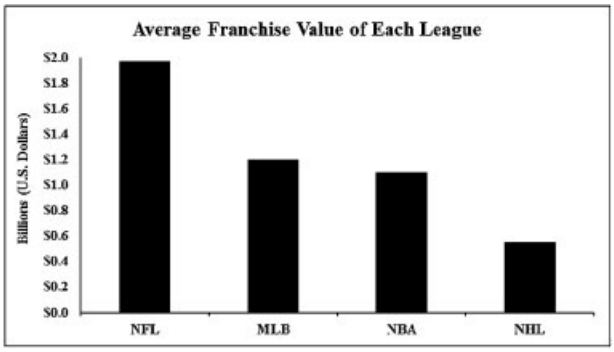

Fig. 2 Average franchise value of the four major American sports leagues (in billions of U.S. dollars).[9–12] MLB, Major League Baseball; NBA, National Basketball Association; NFL, National Football League; NHL, National Hockey League.

National Football League: A Few Sobering Numbers

Despite the financial strength of the NFL and its teams, the economic picture of the league's players is a bit more sobering. At any given time, there are 1,696 active players in the NFL (53 players on each of 32 team's active roster plus a 5-player practice squad), with more than 350 additional players on injured reserve.[13] In 2014, the NFL Players Association (NFLPA) accounted for 1,959 "active players" and 3,130 "former player" members.[14] According to the NFLPA, the average career length of an NFL player is 3.3 years, while the NFL claims that the average is 6 years for players who make an active roster during their rookie season.[15] Injury has been identified as a major predictor of a shortened career span and long-term disability. There were a total of 1,345 reported injuries during the 2014 season, with the league average for Adjusted Games Lost rising from 67.6 in 2013 to 74.3 in 2014,[16] making it 4 years in a row that injury totals around the league increased.

Unlike the other three professional major sports leagues (Major League Baseball, National Hockey League, and National Basketball Association), there are no guaranteed contracts for NFL players. The reason for this is primarily due to the significant number of injuries sustained by the players that necessitates a constant influx of "new" players ("free agents") to take the place of those injured. As a result, there is often a suspicion by players and their agents that team physicians may allow their team's financial situation to influence medical decision making. This is one significant reason that second opinions are so prevalent among NFL players following an injury.

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Similar to any other work-related injury, NFL players receive free health care coverage for injuries sustained during play under worker's compensation. Health care in the NFL is covered under Article 39 of the Collective Bargaining Agreement, which dictates the medical credentials and consultants to be retained by each NFL team:

> Each Club must have a board-certified orthopedic surgeon as one of the club physicians, and all other physicians retained by the Club to treat players shall be board-certified in their field of medical expertise. Each Club will also have at least one board-certified internist, family medicine, or emergency medicine physician (non-operative sports medicine specialist) who must also have a Certificate of Added Qualification (CAQ) in Sports Medicine; any head team physician (orthopedic or medical) hired after the effective date of this Agreement must have a CAQ in Sports Medicine. All Clubs shall have the consultants with the following certifications: neurological (head trauma): board certification in neurosurgery, neurology, sports medicine, emergency medicine, or psychiatry, with extensive experience in mild and moderate brain trauma; cardiovascular: board certified in cardiovascular disease; licensed nutritionist; board-certified neuropsychologist.[17]

All players are allowed to seek a second opinion and receive treatment from the physician of their choice at the team's expense. Vested veterans receive 5 years of continued free health insurance, and the NFL provides financial contributions for eligible former players who need joint replacement surgery. This is especially relevant since an NFL-funded study performed at the University of Michigan in 2009 found that arthritis rates in NFL retirees over the age of 50 years were approximately twice that of the general population with one-in-four ex-players over the age of 50 years having a joint replacement.[18]

Modern Day National Football League Physicians

Historically, the NFL team doctor was an internist, surgeon, or orthopedist who was either the personal physician of the team's owner or relative or close friend. Without the benefit of advanced imaging (i.e., magnetic resonance imaging [MRI]), subspecialty training in sports medicine, or a wealth of clinical and basic science research dealing with athletic injuries, these team physicians were often limited in the expertise they could impart on the sideline and in the clinic. As modern medicine evolved, so did the expertise needed to care for a professional sports team. Today, there are on average, 27 health care providers dedicated to caring for the players during each NFL game, representing several medical subspecialties. For example, in 2013, each team was assigned an unaffiliated neurotrauma consultant stationed on the sideline to assist team physicians in the assessment of potentially concussed players. In addition, an anesthesiologist is also present solely to function as an airway management physician in the event a player requires emergency intubation. In no other aspect of modern society is a young man's injury witnessed in real time (and video recorded from multiple angles) by so many medical specialists who are able to initiate care within a matter of seconds.

Despite this vast wealth of personnel and resources available, caring for an athletic team presents unique challenges for the treating physician. The typical doctor–patient dyad involves the physician working *exclusively* on behalf of the patient with the patient and doctor both sharing the common goal of improving the patient's long-term health in a private setting. On the contrary, professional team medicine is founded on the principle of a physician–patient–team triad where the priority of the athlete is a desire to perform, the patient and doctor both share the goals of supporting immediate athletic achievement, while preserving the patient's long-term health, with multiple parties involved in the relationship. As a result, the team physician often receives input from multiple stakeholders (coaches, general manager, team owner, media, and agents) each with their own agenda. This is coupled with the self-imposed pressure from the physician, himself, related to player approval, being part of the team, financial incentives, job security, and media scrutiny.

The "University of the National Football League"

Coined by John York, MD, owner of the San Francisco Forty-Niners and chairman of the NFL Owners Committee on Health and Safety, the term "University of the NFL" embodies the concept of an interdisciplinary, multispecialty collaboration, utilizing the expertise and medical resources of all health care professionals within a network devoted to making football safer at all levels. The evolution of the NFL physician as a professional entity has been greatly fostered by the NFL Physician Society (NFLPS). Founded in 1966, the NFLPS currently consists of approximately 170 dedicated team physicians and consultants whose mission is to collaborate on solutions to a multitude of injuries and health matters of their patients—the NFL players.[19] Comprised both academic and private practitioners, NFLPS members have completed more than 110 postgraduate fellowships, authored more than 5,500 scientific articles, and care for more than 700 non-NFL sports teams. Members of the NFLPS have published numerous clinical articles in peer-reviewed journals covering a variety of medical issues related to football including the epidemiology of orthopedic injuries,[20–23] prevalence of injury patterns in NFL draftees,[24–26] position-specific injuries,[27] and the influence of various orthopedic conditions on return to play[28] or career length.[29]

The work of the NFLPS is also supported and complimented by several NFL Health and Safety Subcommittees (►**Fig. 3**). These subcommittees, which are funded by the NFL but function independently, are chaired and/or comprised clinicians and basic science researchers who are leaders in their fields, and often not affiliated with an NFL team. Their work has been published in several peer-reviewed journals.

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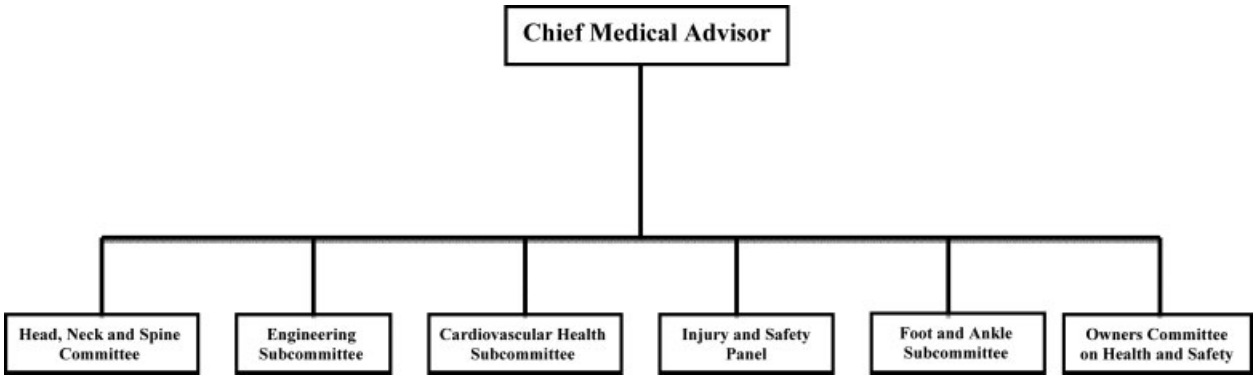

Fig. 3 Medical subcommittees of the "University of the NFL." NFL, National Football League.

Foot and Ankle Subcommittee

The Foot and Ankle Subcommittee has performed multiple studies examining the mechanical interaction between football cleats and playing surfaces,[30] foot kinematics and loading during various football-specific tasks,[31] and the biomechanics of metatarsophalangeal joint ("turf toe") injuries.[32] This last example illustrates the practical application of their in vitro research that showed that most turf toe injuries occur when the first metatarsophalangeal joint dorsiflexes beyond 56 degrees. This information was instrumental in prompting one athletic shoe company (Under Armour, Inc., Baltimore, MD) to bring a football cleat ("V56") to the marketplace that is specifically designed to prevent dorsiflexion of the great toe beyond this injury threshold.

In 2012, the Foot and Ankle Subcommittee was also instrumental in forming the Ad Hoc Committee on NFL field surfaces that formulated "Game Day Surface Guidelines" to improve and standardize the quality of artificial turf and natural playing surfaces with the goal to ultimately diminish the risk of lower extremity injuries. Artificial turf fields are monitored by independent inspectors for various aspects of turf surface hardness and the depth of infill. Stadium field superintendents submit weekly reports to confirm the quality of their fields to ensure compliance.

Engineering Subcommittee

This subcommittee combines a diverse and experienced group of medical and technical experts who study protective equipment to improve player safety. The NFL and NFLPA cosponsored a laboratory study to assess the relative performance of helmets worn by NFL players under laboratory conditions that simulated on-field collisions. A preliminary survey of helmets used by NFL teams was conducted to assess the frequency of use of various helmets in the NFL. Testing involved 17 helmet models intended for use during the 2015 season. All 17 models had achieved National Operating Committee on Standards for Athletic Equipment certification. Under laboratory conditions, the helmets were impacted at eight locations representing the most commonly observed points of contact based on review of NFL game video. Performance during impact testing assessed the effects of rotational velocity and acceleration as there is mounting evidence that concussions mechanistically involve a component of both rotational head moments and direct force vectors. The helmets were then categorized into two groups based on their ability to dissipate the laboratory impact conditions. This information was made available to players and team equipment managers to guide helmet choices.[33]

Toradol Task Force

The "Toradol Task Force" was appointed to deal with the ever-increasing number of players receiving pregame injections of the nonsteroidal anti-inflammatory medication, Toradol (ketorolac tromethamine), before games in an effort to reduce the amount of pain both during and after the game. The concern centered on the increased risk for bleeding a player may have who experiences a traumatic injury after taking ketorolac, which causes an inhibitory effect on platelet function. The task force published its "recommendations" on the safe and appropriate use of this medication based on the manufacturers' recommendations, published side effect profile, and pharmacokinetic data available.[34] This work resulted in a significant reduction in pregame ketorolac injections based on anecdotal reports and an unpublished survey within the NFLPS.

Chief Medical Advisor

In 2015, NFL commissioner, Roger Goodell, appointed Elizabeth Nabel, MD, as the league's first chief health and medical advisor. Dr. Nabel, a cardiologist by training, is a physician-scientist who has published more than 250 academic publications and was the director of the National Heart, Lung, and Blood Institute from 2005 to 2009. At the time of writing, she is the president of Brigham and Women's Hospital in Boston. Her wealth of expertise in appropriating competitive research funding, managing large health care systems, and navigating the vast body of medical literature makes her an ideal candidate to oversee the work of the NFL's various health and safety committees.

Professional Football Athletic Trainers Society

More than 120 certified athletic trainers (ATC) who comprise the Professional Football Athletic Trainers Society (PFATS) are working hand-in-hand with the NFL physicians. Loosely organized in the early 1960s, PFATS was formally established in 1982, working to ensure the highest standard of health care for the NFL players. Athletic trainers are responsible for the day-to-day health care needs of the players, from the on-field

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triage of injured players to their rehabilitation following injury and surgery. PFATS members are often the first line of treatment in the care of injured NFL players.

The National Football League "Combine"

Each February, approximately 320 college football players eligible for the NFL draft are brought to Indianapolis for 4 days to participate in the National Invitational Camp, also known as the Scouting Combine. Beginning in 1982, the Combine's main purpose then, as it is today, was to ascertain medical information on the top draft-eligible prospects in college football. Although medical examinations remain the number one priority of the event, athletes also participate in a variety of psychological and physical tests, as well as formal and informal interviews with team executives, coaches, and scouts from all 32 NFL teams.

Multiple physical metrics are assessed including general physical measurements (i.e., height, weight, hand span, arm length), general strength and speed drills (i.e., 40-yard dash, bench press 225 lb repetitions, vertical jump), and position-specific drills (i.e., passing, blocking, agility). In addition, each player takes the Wonderlic test, a popular 50-item group intelligence test used to assess the aptitude of prospective employees for learning and problem solving.

Each player's medical history is reviewed based on notes from his respective college's athletic training records. The player undergoes a thorough orthopedic exam and is then "presented" to five or six other teams; therefore, a total of six orthopedic exams are administered. Only one internal medicine evaluation is obtained. A full battery of imaging and laboratory tests are available for every conceivable medical problem encountered. For example, at the 2015 Combine, there were 1,123 musculoskeletal X-rays, 508 MRIs, 323 chest X-rays, and 55 computed tomography scans obtained.

Each player is assigned an orthopedic "grade" by each team based on the team's proprietary grading scale that is influenced by the player's injury history and current condition. Though somewhat subjective, this grade is used to facilitate comparison between players while communicating with team administrators. Unlike the normal doctor–patient relationship, the purpose of the medical examinations at the Combine is to assess the player's current ability to play, prove recovery from surgery, gauge the risk for short-term disability, and predict the ability to play beyond the first contract. No consideration is given for a player's long-term health risks. There is only one study, to our knowledge, evaluating the predictive ability of an orthopedic grade at the NFL Combine. Brophy et al[35] reviewed all players medically assessed by the New York Giants from 1987 to 2000 and found that the orthopedic grade assigned at the Combine correlated with the probability of playing in the NFL. While 58% of athletes with a "high" grade and 55% of athletes with a "low" grade played at least one game, only 36% of athletes given a "failing" grade did so ($p < 0.001$). Players with a "high" grade had a mean career of 42 games compared with 34 games for players with a "low" grade and 19 games for a "failing" grade.

Continuity of Care in the National Football League

The clinician's access to clinical documentation—including imaging—at the point of care remains a logistical challenge due to the NFL's dynamic nature, number and variety of injuries, and geographical variables. For example, a player could be injured at a different training venue or in any of the league's 32 stadiums. As a result, imaging studies, in particular, could be generated at multiple locations or imaging centers. Fragmentation of data, lack of timely access, and availability of consultant radiologist review are clearly obstacles to effective point of care delivery. To overcome these obstacles, the NFL instituted a secure, centralized cloud-based picture archiving and communication system (PACS) image repository for all NFL players' electronic images that is securely linked to their NFL identification number and that allows instantaneous, highly automated remote access to imaging at different points of care. Existing studies are preloaded and new images are autorouted to local relay points and uploaded securely to the NFL PACS. All images can be accessed in real time on personal computers or tablets—even from the sidelines—to facilitate diagnosis and treatment. Players' images are automatically linked to their team's active roster based on their unique NFL player identification number and, in case of player movement to another franchise, automatically linked to the new team roster. Once the player leaves his current team, that team's medical staff has no longer access to any of his images.

All imaging is also fully integrated into the NFL's electronic medical record (EMR) system—eClinical Works (Westborough, MA)—which integrates all clinical, pharmaceutical, and imaging data on a player and reflects all treatment rendered regardless of location. The need to have comprehensive medical records available quickly, accurately, privately, and mobile was the driving factor behind the adoption of an EMR system for the NFL. Initiated in 2011 and β-tested in 2013, this system was deployed over the past two seasons and is now in use by all 32 teams.

The "Eye-in-the-Sky"

Starting in 2012, each NFL game has been monitored by a ATC, known as the ATC spotter, located at press box level high above the field of play to oversee those injuries that may have gone unnoticed by the team physicians. This program was initiated following two highly publicized injuries where the players sustained concussions that were not witnessed by their teams' medical staffs. Despite the fact that there are, on average, 27 health care professionals at each NFL game, some injuries are occasionally not seen due to the shear number of players (22) on the field at any one time, numerous players and coaches on the sideline that may obstruct the physicians' view, and the nature of the game whereby multiple injuries can occur simultaneously at locations on the field away from the direction of the play. Once the ATC spotter sees what he interprets as a possible concussion or other significant injury, he immediately calls down to the team's head physician or

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ATC on a secure line to evaluate the player. Not all players whose actions activate this protocol actually sustain a concussion. Occasionally, it becomes evident that the player who was "flagged" has either an orthopedic injury causing him to lose balance or stumble (signs of a possible concussion), or did not actually sustain an injury once he is evaluated. Starting in the 2015 season, the ATC spotter was granted the ability to call down to the sideline official to stop the game—a so-called "Medical Timeout"—so that a player who is still on the field, but possibly concussed, can be taken off and formally evaluated without causing a competitive disadvantage to the player's team. This rule was necessitated by the circumstances experienced by Julian Edelman, a receiver for the New England patriots, who was suspected of sustaining a concussion during the 2015 Super Bowl but remained in the game ultimately scoring a touchdown. There were a total of five "Medical Timeouts" called in the 2015 to 2016 season. It is difficult to determine whether these timeouts were truly "effective" as there are no data available as to whether or not the player would have been medically evaluated in the absence of the "Medical Timeout" or what effects the timeout actually had on player health. Nevertheless, the NFL physicians and athletic trainers have been overwhelmingly supportive of the concept as yet another example of the ways in which the ATC spotter has been an invaluable adjunct to the game day surveillance of injuries.

Game Day Technology

Along with immediate access to players' comprehensive electronic medical history and imaging, and laboratory tests, the NFL has used modern technology in other ways unique to the medical care of athletes. For example, once the ATC spotter identifies a potentially injured player (concussion or other injury with similar signs) and notifies the sideline medical personnel, the injury can be immediately reviewed on a sideline video monitor dedicated solely to injury review. Each play can be seen at normal speed or slow motion, in forward and reverse, and from every angle that it is filmed based on the available network television coverage. Originally instituted to evaluate concussion video, this system has been invaluable in the review of orthopedic injuries to clarify the mechanism and severity of an injury.

Historically, it has been difficult for NFL medical staffs to communicate with each on the sideline given the fact that it spans 100 yards and is occupied by multiple people with various functions. Each ATC and sideline physician are now equipped with a Motorola two-way walkie-talkie (Motorola, Schaumburg, IL). Once an injury is identified or a member of the medical staff is needed to attend to a player, he or she can be immediately contacted via walkie-talkie. In addition, immediate two-way communication is facilitated if a player is taken into the locker room or X-ray room and a status update is needed.

The NFL has also taken advantage of portable tablet technology to facilitate mobile access to the EMR. League-wide use of Microsoft Surface Pro tablets (Microsoft Inc., Redmond, WA) are now used not only for mobile EMR but also in the real-time evaluation of injured players. Each tablet is preloaded with the X2 Biosystems ICE Concussion software (X2 Biosystems, Seattle, WA) dedicated specifically to the assessment of potentially concussed players. The X2 Neurocognitive Software Assessment Tool is an NFL-specific, tablet-based program that incorporates the validated Sport Concussion Assessment Tool (SCAT-3)[36]—an open access instrument that evaluates symptoms, cognitive function, and balance.[37] The X2 adds detailed demographic information from the injury in addition to storing the player's baseline data obtained when he entered the NFL so that injury data can be compared. Completion of the program takes 5 to 7 minutes by an experienced examiner. In use for the past three seasons, the X2 concussion tool has been a very helpful adjunct to evaluate potentially concussed players in a rapid, standardized fashion.

Injury Surveillance

Besides enhancing immediate point of care delivery, EMR technology supports comprehensive injury surveillance and detailed analysis of injury rates and trends. The NFL partners with an outside health care service provider—Quintiles (Quintiles Research Park, NC)—collect and analyze injury data and maintain the EMR database. Quintiles is the world's largest biopharmaceutical development and outsourcing services company whose main tasks are injury data collection and analytics to assess injury trends by player position, mechanism, type of play, and other metrics from practices and games during the preseason, regular season, and playoffs. Injury data are entered into the eClinical Works EMR by each team's medical staff. Quintiles then cross-checks the inputted data against media reports to confirm its accuracy. An injury's circumstances such as game situation are now coded to allow for more detailed root cause analysis and data mining. The data are then analyzed from deidentified EMR data in response to the needs of the NFL, Health and Safety Subcommittees, and NFLPS members. Quintiles submits quarterly injury reports to league administration and each team's medical personnel following their in-season analysis.

As a result of the detail and accuracy afforded by such a comprehensive injury surveillance system, 39 rules changes designed to enhance player safety have been instituted in the NFL since 2005.[33] This, combined with improved equipment design, identification of emerging injury trends and improved injury protocols (i.e., concussion) are all factors leading to improved player health and safety. The fact that players and coaches have embraced a culture change toward injury has also been instrumental in improving player safety. For example, the incidence of concussions has decreased 33% from 252 in 2011 to 202 in 2014 (►**Table 1**). This is felt to be due, in part, to a 68% decrease in hits to defenseless players and a 43% reduction in concussions due to helmet-to-helmet contact since 2012 (►**Table 2**).[38] Recently, there was concern expressed by many that the number of serious knee injuries (i.e., anterior cruciate ligament [ACL] and medial collateral ligament [MCL] sprains) would be increased as a result of the new emphasis on lowering player contact to avoid hits to the head. However, an analysis of the data compiled by Quintiles' analysis determined that the

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Table 1 Incidence of concussions in the NFL[37]

Year	Preseason			Regular season			Preseason + regular season		
	Practice	Game	Total	Practice	Game	Total	Practice	Game	Total
2011	30	48	78	7	167	174	37	215	252
2012	42	43	85	3	173	176	45	216	261
2013	39	38	77	4	148	152	43	186	229
2014	42	41	83	8	111	119	50	152	202
Average	38.3	42.5	80.8	5.5	149.8	155.3	43.8	192.3	236.0

Abbreviation: NFL, National Football League.

Table 2 Concussions by impact source during the regular season: 2012–2014[37]

Impact source for helmet during play	2012	2013	2014
	N	*N*	*N*
Another helmet	91	72	52
Shoulder	25	11	7
Forearm/elbow	4	4	7
Thigh	13	15	15
Foot	2	0	2
Opponent's body (unknown location)	1	2	1
Playing surface	15	25	16
Unknown	13	16	11
Total	173	148	111

number of ACL and MCL injuries was actually lower from 2011 to 2014 (►**Tables 3** and **4**).[38] There was also concern that there would be an increase in the number of injuries that occur during Thursday night games given the decreased recovery time from the prior weekend's game. However, this concern appears to unfounded based on an analysis of the data as there was no significant difference in the number of injuries occurring during Thursday games compared with those on Sunday and Monday over a 4-year period (►**Fig. 4**).[38]

National Football League Sponsored Research Initiatives

In light of the recent attention given to the risk of injury associated with football, the NFL has launched several research initiatives in conjunction with its corporate partners to promote groundbreaking investigation that will hopefully translate into improved player health and safety.[33] In 2012, the NFL launched the Sports and Health Research Program (SHRP) with the Foundation for the National Institutes of Health with a critical imperative to advance scientific understanding of concussion and head injury in athletes, military service members, veterans, and the general population. Uniting diverse areas of expertise in necessary areas of inquiry, the work of these SHRP researchers shows tremendous promise. Thus far, six new pilot-stage research projects have been launched with the goals of improving concussion diagnosis and identifying ways of tracking a patient's recovery.

In 2013, the NFL and GE (General Electric, Fairfield, CT) launched the $60 million head health initiative to stimulate

Table 3 Incidence of anterior cruciate ligament sprains: 2011–2014[37]

Year	Preseason			Regular season			Preseason + regular season		
	Practice	Game	Total	Practice	Game	Total	Practice	Game	Total
2011	9	9	18	4	33	37	13	42	55
2012	15	16	31	3	29	32	18	45	63
2013	15	12	27	3	27	30	18	39	57
2014	12	10	22	0	27	27	12	37	49
Average	12.8	11.8	24.5	2.5	30.5	33.0	15.3	42.3	57.5

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Table 4 Incidence of medial collateral ligament sprains: 2011–2014[37]

Year	Preseason			Regular season			Preseason + regular season		
	Practice	Game	Total	Practice	Game	Total	Practice	Game	Total
2011	9	27	36	4	103	107	13	130	143
2012	19	26	45	9	78	87	28	104	132
2013	20	27	47	4	85	89	24	112	136
2014	16	22	38	3	97	100	19	119	138
Average	16.0	25.5	41.5	5.0	90.8	95.8	21.0	116.3	137.3

research and development around new brain imaging technologies,[33] and divided into three challenge grants: challenge I is dedicated to identifying new tools for concussion diagnosis; challenge II, in partnership with GE and Under Armour (Baltimore, MD), will be devoted to improving the detection and prevention of brain injury; and challenge III, assisted by the expertise of the National Institute of Standards Technology will explore the development of new synthetic materials that may enhance player safety. Thus far, more than 1,000 concepts have been submitted from 30 countries. The work has achieved promising breakthroughs, from blood tests that speed the diagnosis of brain injury to new solutions for mitigating the force of impact. Clearly, the work that is developed under these research grants may be used by other entities, such as the Department of Defense, to improve the health and safety of their service members.[39]

Conclusion

Despite the fact that American football will always possess some degree of injury risk due to the nature of the game, the size and strength of its players, and the speed and forces exerted during play, the NFL and its medical partners have made significant strides to improve injury surveillance, treatment, and prevention. Significant efforts have been made by multiple stakeholders to use advances in digital health care technology and incorporate them into a comprehensive, secure, and seamless league-wide program. The "University of the NFL" comprised NFL team physicians, outside clinicians, and researchers have combined their talents with the financial support of the NFL to provide the most advanced health care of any sports league. The results of this work will undoubtedly influence not only NFL players but also athletes in general, and football players, in particular, at all levels of play. Other professional and/or collegiate sports leagues should strongly consider the model of the "University of the NFL" as a means of improving the health and safety of its athletes.

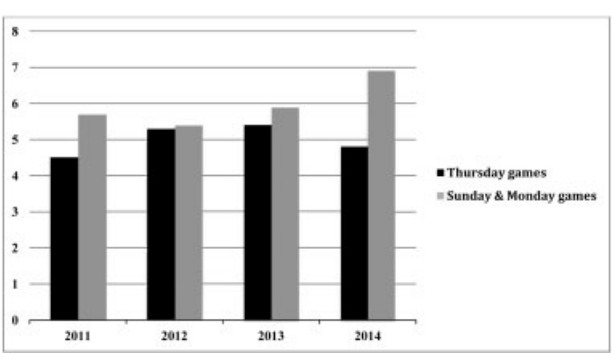

Fig. 4 Comparative injuries on Thursday night games compared with Sunday and Monday games.[37] There was no significant difference in the number of injuries occurring on Sunday and Monday games compared with Thursday games.

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